

DR. ANJALI DHABU

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Date of birth: October 5th, 1989

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● EDUCATION

- 2014 - 2020 Ph.D. in Civil Engineering
Division of Structural Engineering, Department of Civil Engineering,
Indian Institute of Technology (IIT), Madras, India
Advisor: **Prof. Dr. S.T.G Raghukanth**
- 2012 - 2014 M.Tech, Structural Dynamics and Earthquake Engineering
Department of Applied Mechanics,
Visvesvaraya National Institute of Technology (VNIT), Nagpur, India
Advisor: Prof. Dr. O.R.Jaiswal
- 2007 - 2011 B.E, Civil Engineering
Department of Applied Mechanics and Civil Engineering,
Shri Govindaram Seksaria Institute of Technology, Indore, India

● CURRENT POSITION

- 04/2024 – 03/2027 Postdoctoral Researcher, **Individual Research Grant funded by DFG**
Project: Heterogeneities and its Effect on Rotational Seismology (HERS)
Institute of Geophysics, Faculty for Mathematics, Computer Science and Natural
Sciences, University of Hamburg, Germany

● PREVIOUS POSITIONS

- 07/2023 - 03/2024 Postdoctoral Researcher
Project: ErUM-WAVE: Anticipation of 3-dimensional wave-fields.
Supervisor: **Prof. Dr. Céline Hadziioannou**
Institute of Geophysics, Faculty for Mathematics, Computer Science and Natural
Sciences, University of Hamburg, Germany
- 11/2021 - 08/2023 Postdoctoral Researcher
Project: Building vibrations: combined condition analysis with innovative Sensor
concept (GIOTTO)
Supervisor: **Prof. Dr. Céline Hadziioannou**
Institute of Geophysics, Faculty for Mathematics, Computer Science and Natural
Sciences, University of Hamburg, Germany

● PROJECTS

- 341,497€ Individual Project funded by Deutsche Forschungsgemeinschaft (DFG)
Title : Heterogeneities and its Effect on Rotational Seismology (HERS)

● PUBLICATIONS

► Journal Publications

1. **Dhabu, A.C.**, Varun K. Singla & Raghukanth, S.T.G. Seismic wave propagation through layered reduced micropolar medium, Journal of Geophysical Research: Solid Earth, (2021) 126, e2020JB020931. <https://doi.org/10.1029/2020JB020931>
2. **Dhabu, A.C.**, & Raghukanth, S. T. G. Translational and rotational ground motion simulations in homogeneous reduced micropolar half-space, Journal of Seismology, (2021), 25, 599–623.

<https://doi.org/10.1007/s10950-021-09983-2>

3. **Dhabu, A.C.** & Raghukanth, S.T.G. Fundamental Solutions to Static and Dynamic Loads for Homogeneous Reduced Micropolar Half-Space, Pure and Applied Geophysics. (2019) 176 (11): 4881-4905. <https://doi.org/10.1007/s00024-019-02225-0>
4. **Dhabu, A.C.** & Raghukanth, S.T.G. Influence of Himalayan topography on earthquake ground motions, Arabian Journal of Geosciences, (2021), 14, 1931. <https://doi.org/10.1007/s12517-021-08111-1>
5. **Dhabu, A.C.**, Sugumar, S. & Raghukanth, S.T.G. Characterization of Strong Motion Generation Regions of Earthquake Slip Using Extreme Value Theory, Pure and Applied Geophysics. (2019) 176 (8): 3567-3592. <https://doi.org/10.1007/s00024-019-02136-0>.
6. Kumar, P. S., Mohanty, R., Lakshmi, K. J. P., Raghukanth, S. T. G., **Dhabu, A. C.**, Rajasekhar, R. P., & Menon, R. (2019). The seismically active lobate scarps and co-seismic lunar boulder avalanches triggered by 3rd January 1975 (Mw 4.1) shallow moonquake. Geophysical Research Letters, 46(14), 7972-7981. <https://doi.org/10.1029/2019GL083580>.
7. Kumar, P. S., Krishna, N., Lakshmi, K. P., Raghukanth, S. T. G., **Dhabu, A.**, & Platz, T. (2019). Recent seismicity in Valles Marineris, Mars: Insights from young faults, landslides, boulder falls and possible mudvolcanoes. Earth and Planetary Science Letters, 505, 51-64. <https://doi.org/10.1016/j.epsl.2018.10.008>.
- **Manuscripts under review and under preparations**
8. **Dhabu, A.C.**, Bernauer, F., Liao, C.M., Niederleithinger, E., Igel, H. & Hadziioannou, C., Characterizing Rotational Ground Motions: Implications for Earthquake-Resistant Design of Bridge Structures (2024). *arXiv preprint arXiv:2411.02203*.
9. Hejazi Nooghabi, A., **Dhabu, A.C.**, Montellier, V. & Hadziioannou, C.: Development of coupled ground motion simulation technique for teleseismic events (to be submitted to Geophysical Journal International)
10. **Dhabu, A.C.**, Matthießen, N., & Hadziioannou, C., Parametric analysis of heterogeneities and their effect on simulated rotational ground motions and strains due to earthquakes (to be submitted to Bulletin of Seismological Society of America)

● INVITED TALKS AND CONFERENCE PRESENTATIONS

- Müller, L., **Dhabu, A.**, Bernauer, F., Donner, S., Bode, K. and Hadziioannou, C., 6-Component Operational Modal Analysis of wind turbines for damage detection, *13th International Conference on Structural Health Monitoring of Intelligent Infrastructure (SHMII)*, Graz, Sept.1-5, 2025 <https://doi.org/10.3217/978-3-99161-057-1-120>
- Hejazi Nooghabi, A., **Dhabu, A.**, and Hadziioannou, C.: Developing a High-Resolution Subsurface Model through Teleseismic Wave Simulation in Hamburg, *EGU General Assembly 2025*, Vienna, Austria, 27 Apr–2 May 2025, EGU25-18687, <https://doi.org/10.5194/egusphere-egu25-18687>, 2025.
- F. Bernauer, P. Bonkowski, **A. Dhabu**, Z. Zembaty. Applications of Rotational Sensing in Analyzing Structural Vibrations and their Modal Analyses, *Keynote speech at 18th World Conference in Earthquake Engineering (WCEE)*, Milan, Italy, 2024
- Bernauer, F., Balaskas, G., Hadziioannou, C., **Dhabu, A.**, Liao, C.M., Niederleithinger, E., Yuan, S., Wassermann, J. and Igel, H., (2023), December. Seismic monitoring of civil engineering structures with 6C measurements. In *AGU Fall Meeting Abstracts* (Vol. 2023, pp. S54C-06).
- **Dhabu, A.**, Bernauer, F., Liao, C.M., Hadziioannou, C., Igel, H. and Niederleithinger, E., (2023), May. Using Rotational Motions to understand material damage in Civil Engineering structure. In *EGU General Assembly Conference Abstracts* (pp. EGU-8851).
- Invited talk at Department of Earth and Environmental Sciences, *Ludwig-Maximilians-University* on the topic, 'Modal analysis of simplified bridge structure using 6C recorded data' 2023, Munich, Germany
- **Dhabu, A.C.** & Raghukanth, S. T. G. Reduced micropolar theory: an alternative to model the Earth medium and simulate earthquake ground motions? *6th Conference of International Working Group on Rotational Seismology (IWGoRS)*, 2022, Paris, France
- *Invited talk* on the topic 'Simulation of Translation and Rotational Ground motions' at *SPIN-ITN workshop: Physics and Dynamic Processes* held at Carcans, near Bordeaux, France from May22–28,2022.
- Presented a poster 'Seismic Wave Propagation in Reduced Micropolar Half-space' at *SPIN-ITN workshop: Physics and Dynamic Processes* held at Carcans, near Bordeaux, France from May 22–28,2022.

- **Dhabu, A.C.**, Varun K. Singla & Raghukanth, S. T. G. Analytical formulation and numerical simulations for layered reduced micropolar medium subjected to double-couple forces. *5th Conference of International Working Group on Rotational Seismology (IWGoRS)*, 2019, Taipei, Taiwan
- MohammedAtif, **Dhabu, A.C.**, Raghukanth S.T.G, & Manam S.R. Response due to a buried source in reduced micropolar elastic half-space. *5th Conference of International Working Group on Rotational Seismology (IWGoRS)*, 2019, Taipei, Taiwan
- **Dhabu, A.C.** & Raghukanth, S. T. G. Influence of Himalayan Topography on Earthquake Strong Ground Motions. *1st Conference of the Arabian Journal of Geosciences (AJGS)*, 2018, Hammamet, Tunisia. https://doi.org/10.1007/978-3-030-01656-2_39.
- **Dhabu A.**, Dhanya J., & Raghukanth S.T.G. Effect of Topography on Earthquake Ground Motions. *Structural Engineering Convention (SEC)*, 2016, Chennai, India. https://doi.org/10.1007/978-981-13-0365-4_9.

● EXPERIENCE

► Teaching Experience

- *Co-supervision of M.Sc Student*, Mr. Laurin Müller at the University of Hamburg for his Masters thesis titled ‘Structural Health Monitoring of wind turbines using 6C ground motion data’.
- *Co-supervised B.Sc Student*, Mr. Nicolas Matthießen at the University of Hamburg for his Bachelor thesis titled ‘Numerical simulation of seismic gradients and their response to heterogeneity in the Earth medium’.
- Taught online course on SPEC-FEM, a community solver for seismic wave propagation based on the Spectral Element Method in SPIN-ITN short course on Computational Seismology.
- *Co-Supervised three Dual degree/M.tech students*, Mr. Kandarp Pathak, Mr. Akhil Manikala, and Mr. Jaifer Muhammed registered with Prof. Dr. S T G Raghukanth, for their final year Master’s projects.
- Assisted Prof. Dr. S T G Raghukanth for making assignments, conducting tutorials, setting question papers and their evaluation of multiple civil engineering subjects at IITM namely, Structural Dynamics, Design of steel structures, Stability of Structures, Probability, Advanced Mechanics of Structures and Introduction to Research.
- Assisted Prof. Dr. O.R. Jaiswal for assignments and tutorial sessions for undergraduate Engineering Mechanics course at VNIT.

► Research Experience

- Reviewer for *Geophysical Journal International* and *Seismica*.
- Reviewer (2024–2026) on the panel of the *Ideas and Venture Fund (IVF)*, a competitive internal funding scheme at the University of Hamburg.
- Co-organizer, *AG-Seismologie 2024* Conference, University of Hamburg, fostering scientific exchange alongside Prof. Dr. Céline Hadziioannou and Prof. Dr. Conny Hammer.
- Co-organizer, short courses and workshops for *SPIN-ITN* PhD researchers and external candidates, promoting collaborative research and skill development (with Prof. Dr. Céline Hadziioannou).
- Project Associate, Indian Institute of Technology Madras (IITM), to analyze Seismic Wave Propagation on the Moon and Mars (6 months).
- Research Assistant, with Dr. S. T. G. Raghukanth, to assess the vulnerability of Public Health Facilities in coastal Odisha, focusing on structural and non-structural components.

● ACADEMIC AWARDS

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| 2020 | - <i>Institute Research Award</i> , IIT Madras, India, in recognition of the quality and quantity of research output during Ph.D. studies. |
| 2014 | - <i>Silver Medal</i> , VNIT, Nagpur, India — for securing second position in the Department of Applied Mechanics during the M.Tech programme. |
| 2011 | - <i>Prof. D. G. Dhavalikar Memorial Gold Medal</i> , <i>Shri K. G. Seksaria Memorial Medal</i> , <i>Planarch Gold Medal</i> , <i>Prof. B. S. Phadke Award</i> , and two additional <i>Gold Medals</i> , for achieving highest overall marks, first position throughout the course and in the final year, and highest marks in Structural and Geotechnical Engineering during the B.E. (Civil) programme at Shri G. S. Institute of Technology and Science, Indore, India. |

2010 - *Gold Medal*, Shri G. S. Institute of Technology and Science, Indore, India, for securing the **highest marks in Geotechnical Engineering** during the B.E. (Civil) programme.

Career breaks, diverse career paths and major life events:

June 2020 – November 2021: Following the completion of Ph.D. in India, I intended to pursue a research career abroad. However, international mobility was severely restricted due to the COVID-19 pandemic. During this period, I also took on significant family responsibilities, caring for my mother-in-law, who passed away in June 2021. After fulfilling these obligations and as global travel restrictions eased, I resumed the academic trajectory and began my first postdoctoral position at the University of Hamburg with *Prof. Dr. Céline Hadziioannou* in November 2021.

Research Expertise, Leadership, and Vision:

I am a trained **civil engineer** specializing in **Structural Dynamics and Earthquake Engineering**, with over a decade of **research experience in seismic wave propagation and rotational seismology**. Among the few researchers worldwide to have applied Reduced Micropolar Theory (RMPT), I bridge seismology and civil engineering by translating complex theoretical concepts into practical frameworks for real-world engineering challenges. Having completed my entire education in India as a top-performing student with multiple academic awards, I later advanced my research career at the University of Hamburg, Germany, successfully integrating into a diverse academic and cultural environment. My **work employs advanced analytical and numerical models to represent the Earth's medium, linking theoretical seismology with earthquake engineering applications**. My **background in civil engineering plays a vital role in my research on structural health monitoring**, enabling me to **connect seismic modeling insights directly with the behavior and resilience of built structures**. I have demonstrated strong research independence through my **DFG-funded individual project *HERS*** and through active engagement with the international scientific community via conferences, workshops, peer review, and event organization. With a collaborative and mentoring-focused approach, I strive to cultivate a dynamic research environment that promotes innovation, critical thinking, and scientific creativity. My long-term vision is to establish a leading research program in seismic wave-propagation modeling, advancing the understanding of rotational ground motions and developing **innovative strategies for structural health monitoring and seismic hazard mitigation**, while guiding and inspiring the next generation of scientific leaders.